KATHRYN THOMPSON

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EXPERIENCE

Construction Engineering Research Laboratory, USACE – *Champaign, IL* *June 2024 – Present*
Computer Scientist (Sept. 2025 – Present); Cyber Resilience Engineering Intern (June 2024 – Sept. 2025)

- Tested, documented, and successfully demonstrated Control system Agnostic Secure Enclave (CASE) across 8 buildings at 2 sites, validating secure communication, installation, and scalability while achieving technology costs of just 10% of the vendor alternative.
- Developed and automated ROS2 wireless communication, resolving compatibility issues and strengthening secure device-to-device communication while adhering to safety requirements.
- Defined standardized test plans for building control systems to evaluate functionality, interoperability, maintainability, and cybersecurity, establishing consistent testing procedures across laboratory capabilities.
- Built Shiny for Python data visualizations to support migration from Tableau to open-source alternatives, improving user experience and generating thousands in software licensing savings.
- Contributed to UFGS specification development by defining construction scopes of work, technical requirements, and regulatory standards for building-control projects.
- Presented laboratory and project capabilities to government, academic, and industry stakeholders through conferences and technical tours, communicating project impact to support funding, collaboration, and technology adoption.

School of Information Sciences, University of Illinois – *Champaign, IL* *Oct. 2024 – May 2025*
Undergraduate Cybersecurity Research Assistant

- Conducted literature review on post-quantum cryptography (PQC) adoption in operational technology (OT), analyzing latency, encryption overhead, and regulatory compliance in systems including ROS2.
- Led vulnerability testing for security and data privacy of healthcare-facing applications.

National Aerospace Solutions – *Arnold AFB, TN* *May 2023 – Aug. 2023*
Electrical and Mechatronics Engineering Intern

- Modified C code to resolve communication errors between a PIC microcontroller and DHCP server, including incorrect IP addresses, MAC addresses, and hostnames.
- Updated 13 technical manuals to current operating procedures and ASME standards and designed a Microsoft Access status board consolidating 7 plant equipment databases.

EDUCATION

University of Illinois Urbana-Champaign – *Champaign, IL*
M.S. Computer Science *Aug. 2025 – Aug. 2026*
B.S. Computer Engineering *Aug. 2021 – May 2025*

Leadership: ACM Member; ECE Student Advancement Committee President (2023–2025), Academic Chair (2022–2023); Broadening Participation in ECE Undergraduate Representative (2023–2025)

Certifications and Awards: Illinois Trust Institute Certificate in Cybersecurity, CyberCorps SFS Recipient

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, JavaScript, HTML, Bash, SystemVerilog, Assembly, G-Code, LC-3
Frameworks/Libraries: ROS2, Shiny, Dash, NumPy, Pandas, SciPy, Matplotlib, Plotly, ScaPy, Electron
Tools: Linux, Fedora, Docker, Git, Wireshark, SpecsIntact, Cisco Packet Tracer, Microsoft Access, PowerApps, Android Studio, Autodesk Inventor